

Project presentation on

“Smart Movable Divider for Ambulance Path Optimization ”



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Academic Year: 2024-2025

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Introduction

The Smart Movable Divider for Ambulance Path Optimization system is designed to reduce ambulance response times in congested areas by creating clear, dedicated lanes using IoT-enabled movable dividers and AI-based traffic management. This system dynamically adjusts traffic flow, prioritizing ambulances in real-time to ensure faster, more efficient emergency responses and potentially save lives.

Problem Statement

Develop a smart movable divider system for ambulances that optimizes their path through urban environments, reducing response times and improving accessibility. The system should dynamically adjust to traffic conditions and obstacles, ensuring efficient navigation to emergencies.

Goals And Objectives

- Reduce Response Time: Minimize ambulance delays in traffic.
- Real-Time Traffic Monitoring: Track live traffic for instant decision-making.
- AI-Based Route Optimization: Use AI to find the fastest ambulance route.
- Dynamic Lane Management: Create emergency lanes with movable dividers

Solution

- We created a smart movable road divider
- Camera Detect Ambulance in Traffic
- The Detection Signal Send To the Arduino System through Serial Communication
- The Arduino Move motor to Shift Divider
- After Ambulance Pass Divider come to their original position

Literature Survey

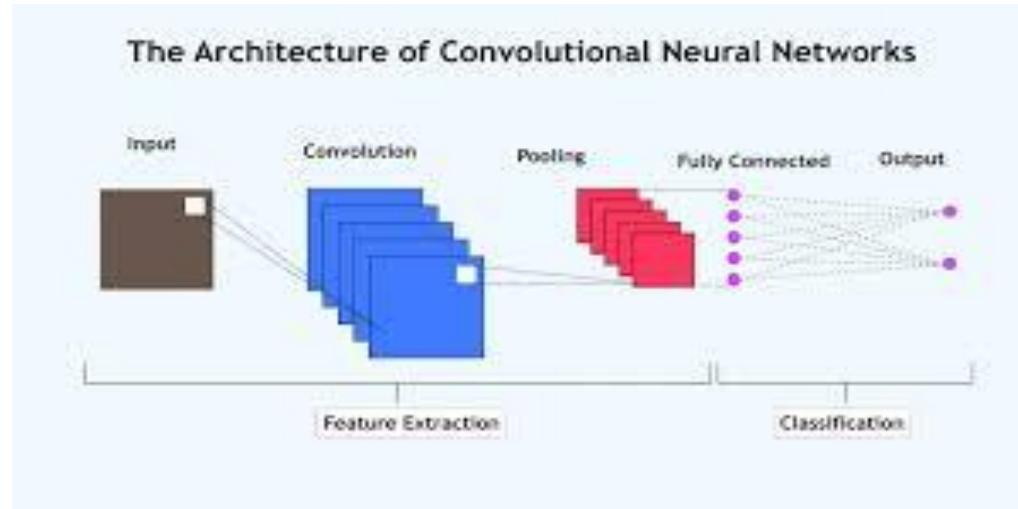


Name	Author	Year	Description	Drawback
CONTROLLING OF SMART ROAD DIVIDER FOR CLEARANCE OF AMBULANCE PATH	Monoj H. M., N. Sangita,	2024	This project proposes a dynamic, IoT-powered movable road divider that adapts to real-time traffic conditions using computer vision and deep learning. By shifting lanes during peak hours, it optimizes traffic flow, prioritizes emergency vehicles, and enhances road safety.	1. High implementation cost due to advanced technology and infrastructure changes 2. Potential technical malfunctions or inaccuracies in traffic density detection
“SMART MOVABLE ROAD DIVIDER	Adity S. Gunjal, Shubham V. Erende	2024	The "Smart Movable Road Divider for Clearance Ambulance Path and Traffic Control" project aims to create a dynamic road divider system that adjusts in real-time to optimize traffic flow and prioritize ambulance clearance.	1. High maintenance requirements for the advanced technology infrastructure.

3.MOVABLE ROAD DIVIDER FOR VEHICULAR TRAFFIC CONTROL WITH MONITORING OVER	Mrs. Anita Chowdhary, M. Ganesh,	2023	The "Movable Road Divider for Vehicular Traffic Control with Monitoring" project focuses on creating a flexible road divider system that adjusts in real-time based on traffic conditions. It leverages IoT and monitoring systems to optimize traffic flow, enhancing overall road efficiency and safety.	1. High setup and operational costs for implementing real-time monitoring and control systems 2. Risk of system failure or inaccuracies in traffic detection, affecting performance
4.AUTOMATIC MOVABLE DIVIDER FOR TRAFFIC MANAGEMENT[	G. Mansa, M. Kalpana, J. Praniti	2023	The "Automatic Movable Divider for Traffic Management" project aims to develop a dynamic road divider that automatically adjusts to traffic patterns. Utilizing real-time data and automation, the system helps to optimize traffic flow, reduce congestion, and enhance safety.	1. Significant investment required for integrating automation technology and sensors 2. Potential issues with system reliability in harsh weather or unexpected

Algorithms/Techniques Used

- CNN algorithm used For Image Feature Extraction And Detection



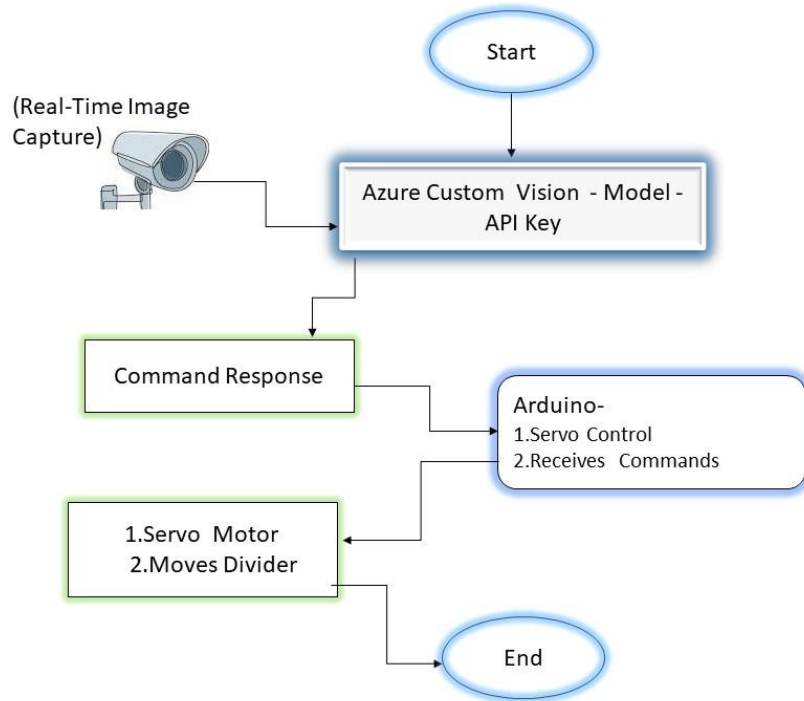
Modules Split-up

- 1.Real Time Image Capturing- Capture Image From the Camera
2. Yolo model detect Ambulance
- 3.Sending Command To Arduino
4. Servo Motor Move Divider

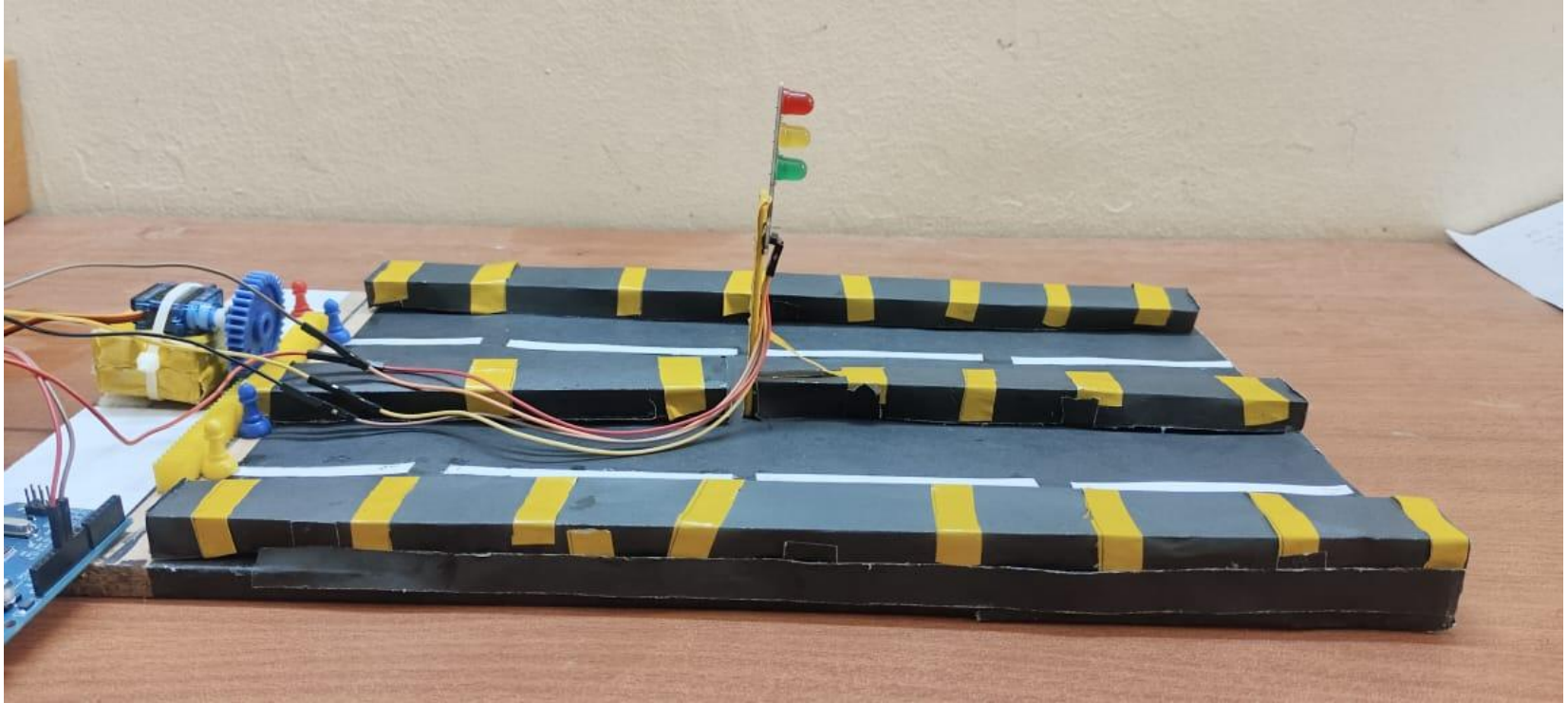
Software Tools/Hardware tools

- jupyter notebook
- Arduino IDE
- Vs code
- Servo
- Arduino
- Jumper wires

System Architecture



Result



Conclusion

The proposed Smart Movable Divider System optimizes ambulance response times by dynamically creating clear paths using real-time traffic data and AI-driven control. This system enhances emergency efficiency, reducing delays and potentially saving more lives

Advantages and Applications

Advantages:-

- 1.Reduced Response Time – Optimizes ambulance routes, ensuring quicker medical intervention.
- 2.Dynamic Traffic Management – Real-time lane adjustments reduce congestion and improve flow

Applications;-

- 1.Emergency Medical Services– Ensures faster response times for ambulances in cities.
- 2.Smart Cities– Enhances traffic management in urban areas with high congestion.

Future Scope

- In the future, this system can be integrated with autonomous vehicle technologies to further streamline emergency responses.
- expanding the system to include predictive traffic analysis could enhance its effectiveness in anticipating and preventing congestion in high-traffic areas.

References

1. Singh, R., Gupta, P. (2020). Ambulance Response Time Optimization Using Dynamic Lane Management. *Journal of Intelligent Transportation Systems*.
2. Kumar, A., Bansal, M. (2019). Smart Traffic Management for Ambulance Using IoT and AI-based Road Dividers. *International Journal of Advanced Research in Computer Science*.
3. Jha, S., Mishra, V. (2021). Emergency Vehicle Priority with Movable Dividers and Smart Traffic Lights. *IEEE Transactions on Vehicular Technology*.

Thank You

